

ARI 410/510 Lab 2: Unsupervised Learning and Feature Reduction

University of Michigan-Flint

Fall 2026

(See course Canvas page for due date. Post questions on Discord in the #homework-help channel.)

1 Overview

The goal of this assignment is to introduce you to unsupervised learning techniques, including dimensionality reduction and clustering, and to explore how these techniques relate to choosing which features a model should use. You will work with the **Human Activity Recognition Using Smartphones Dataset**, which contains measurements from smartphone sensors (accelerometers and gyroscopes) recorded while participants performed six activities (walking, walking upstairs, walking downstairs, sitting, standing, and lying down).

The dataset consists of 561 features extracted from the raw sensor data. You will explore the dataset, reduce its dimensionality using **Principal Component Analysis (PCA)**, perform clustering, and compare two different ways of shrinking the feature set before training a classifier. By the end of this assignment, you should be able to say how much a model gives up (or doesn't) when it uses a fraction of the original features, and what you give up or gain depending on *how* you reduce the feature set.

The dataset can be downloaded from the UCI Machine Learning Repository. A short starter notebook (on Canvas) handles downloading and loading the files, including attaching the activity names to the labels, and includes a few general-purpose plotting helpers (shown on toy data) that you can use for your own figures. Everything else is up to you.

Read this before you start: if you use any generative AI on this assignment, you will fill out a short structured statement about how you used it (Section 6). Attaching your chat logs is optional but encouraged; if you plan to do that, set up whatever you need to export them *before* you begin.

2 Instructions

Step 1: Setup and standardization

Split off part of the provided training set as a **development (dev) set**. As in Lab 1, use the dev set for comparisons along the way, and use the provided **test set** only once, at the very end. Standardize the features so that each has a mean of 0 and a standard deviation of 1 (e.g., with sklearn's StandardScaler); PCA and clustering work better with standardized data. Think about which data each of these steps should be *fit* on, and which should only be *transformed* using that scaler. You don't need to report anything from this step in the submission.

Step 2: Exploratory Data Analysis (EDA)

Perform an initial exploratory analysis of the dataset. This is always a good idea with a new dataset, so you get a feel for what you are working with before you train any models. There is some code in the starter notebook with examples of how to create basic plots in Python.

- Visualize the distribution of the activities (e.g., a histogram or bar chart). Use the activity names in your plot (e.g., "Sitting"), not the integer codes.

- Apply **PCA** to the features and plot the **explained variance ratio** for each principal component (see `explained_variance_ratio_` in the sklearn PCA class). Identify the “elbow” of this plot, where the values level out and stop decreasing sharply. The number of components at the elbow is your value of k ; keep it in mind for later.
- Visualize the data as a 2D scatter plot using the first two principal components, colored by activity.

Step 3: Clustering

- Apply **K-Means clustering** to the features and use the **silhouette score** to choose the number of clusters (e.g., try 2 through 10). Visualize the clusters using the first two principal components, colored by cluster this time. How do the clusters compare to the activities?
- Repeat the clustering analysis using **DBSCAN** and compare the results to K-Means. DBSCAN does not take a number of clusters as a hyperparameter, but its results depend heavily on its `eps` hyperparameter, so **you will need to experiment with eps**.

Step 4: Two ways to shrink the feature set

- **Transform:** use the top principal components from PCA. Note that these are *new* features, each a combination of the original 561.
- **Select:** train a random forest and use its `feature_importances_` to keep only the most important of the original features. Look at which features these are. Note that this approach is just using the random forest scores to select features, it can be used together with random forest as a classifier or these features can be used for a different classifier altogether (see next step).

Step 5: Supervised learning with fewer features

- Train a simple classifier (e.g., logistic regression or SVM from sklearn) on the full feature set, and on reduced feature sets from each approach in Step 4.
- Your elbow k from Step 2 is one choice for how many dimensions to keep, but it is not the only one. Explore how performance on the dev set changes as you keep more or fewer dimensions with each approach, and present what you find in a plot.
- Finally, evaluate the full model and your chosen reduced model(s) on the **test set** (the first time you use it!) using accuracy, precision, recall, and F1-score.
- Please see the report section for specific requirements for ARI 410 and 510 related to this step.

3 Deliverables

You should submit:

- a written report (PDF), and
- the code used for your experiments. A Jupyter notebook is completely fine; if you submit multiple files, include a short README explaining how to run them.

If you want to combine these, you can submit a PDF export of a Jupyter notebook that mixes code, visualizations, and text, as long as the report sections below are clearly labeled. You may also put your report in a GitHub README.md with links to the code in the same repository.

4 Report

The report should include the following sections. With each figure, write a brief description of what it tells us about *this* dataset (i.e., do not just say “a histogram shows the distribution of the data”; talk about what *this* histogram tells us about *this* dataset).

Everyone

1. **Exploratory Analysis:** the activity distribution, the explained variance plot, and the 2D PCA plot colored by activity, each with a description. Include your value of k and how you chose it.
2. **Clustering:** your silhouette analysis and at least 2 visualizations (one for K-Means and one for DBSCAN). Describe what you observed with each method, including how the clusters relate to the activities and what happened as you varied ϵ .
3. **Classification:** your results for the full feature set and the reduced feature sets, including your exploration of how many dimensions to keep. Reflect on the results. How did reducing the number of features affect performance, interpretability, and efficiency? How did the two approaches compare, and why do you think that is?
4. **Reflection:** a paragraph on the overall process. You might answer questions like: what was the most challenging part? How did you overcome it? What is your main takeaway? Did anything surprise you?

ARI 410, additionally

1. In step 5, which activities get confused? Create confusion matrices on the dev set for at least one reduced model and the full model: what gets mixed up, what stays the same in both cases?

ARI 510, additionally

1. In step 5, also consider efficiency: what does using fewer features actually save? Measure something relevant (e.g., training or prediction time, memory usage, etc.) and present that measurement in a figure or table, and write about what else might matter.

ARI 410 students are welcome to include this as well.

5 Code

Make sure the code is organized and readable, so that it is easy to see which parts correspond to each part of the assignment. If you used any non-standard libraries (other than core Python, sklearn, or common data visualization libraries), note how to install them.

6 Generative AI Use Statement (required)

Every submission must include **one** of the following. This requirement is the same as the previous labs.

Option A: No AI. If you used no generative AI, state exactly: “*No generative AI was used during this assignment.*” You are done with this section. Please note that typing a query into Google and getting the automatic Gemini response (or something similar to that, which I will call “unintentional AI use”) doesn’t need to be documented here. Only include cases where you actively used a GenAI tool to help in some way (including autocomplete for coding, ChatGPT chat interface, agentic assistant like Claude Code, etc.).

Option B: Structured statement. If you used any generative AI tool you deliberately turned to (ChatGPT, Claude, Copilot, Gemini as a chatbot, etc.), complete the table below with one row per phase of your work. The point is not to confess to using AI (it is allowed) but to show *where the thinking was*: research on

knowledge work finds that when people use AI, the real effort shifts from producing an artifact to verifying, integrating, and deciding whether to trust the output.¹

For each phase, note: (1) whether you used AI, and if the answer is “yes”, fill in: (2) *how*, i.e., which mode(s) (see below), and what you actually asked (feel free to make this cell of the table large to explain) and (3) what you did to check, correct, or override the output. The three “modes”:

- **Creation:** had the AI generate something from scratch (code, text, an idea).
- **Information:** I used it to explain, find, or summarize (e.g., “what does the SVM C parameter do?”).
- **Advice / feedback:** I used it to improve, check, or critique something you already had (e.g., “review this function,” “is F1 the right metric here?”).

Phase	Used AI?	How (mode + what you asked)	How you checked / corrected / overrode it
Planning & understanding			
Coding and Debugging			
Analysis & interpretation			
Report writing, formatting, editing			

[Optional, but encouraged] **Attach your logs.** For any deliberate chatbot or AI agent use, export and attach your chat transcript or coding session, or include a shared link. Set this up *before* you start (this is why it’s flagged in the Overview). If you used an inline tool that cannot export a clean log (e.g., editor autocomplete), just describe how you used it instead.

7 Other questions?

Please feel free to ask on Discord in the [#homework-help](#) channel.

¹Structure adapted from H.-P. Lee, A. Sarkar, L. Tankelevitch, I. Drosos, S. Rintel, R. Banks, & N. Wilson, “The Impact of Generative AI on Critical Thinking,” CHI 2025.